Bounded scores and well-specified families: when the paper's contrastive objective works, and when the loss variant matters

Abstract

The goals 1–3 write-up (`docs/goals1-3_writeup.pdf`) ended with two empirical patterns. First, every catastrophic failure was an unbounded score on heavy-tailed or spiky data. Second, the paper's sharpest predictions (the dot product's wrong spread, P1, and the joint loss's shrinkage, P4) held for linear encoders and disappeared for MLPs. This note gives both patterns a theoretical footing inside the paper's own framework [1]. *Part I (boundedness):* a score that is bounded above defines a valid model for every data distribution, and its bank forecast is consistent. A score that is linear in the future does not: on Student-t futures its model does not exist for any informative parameter, ReLU MLPs do not change this, and the bank forecast provably collapses onto a single future as the bank grows. *Part II (misspecification):* the one-way, two-way and joint losses have the same population target, the log density ratio $\log r$, up to different free terms. So they agree whenever the score family can represent that target, and can differ only when it cannot. In the Gaussian case this recovers P1, P3 and P4 as instances of misspecification. It also gives a new result: linear L^2 trained two-way or joint is misspecified on any correlated Gaussian. At weak dependence it loses up to 3/8 of the available information, yet only 0.3% on the goal-1 field, which is why goal 1 did not see it. Every numerical claim is reproduced by `test_bed/theory/misspecification.py`.

1 Setting and notation

We follow the paper [1]. The past is $u \in U$, the future is $v \in V$, the data distribution is $\mu(du, dv)$, and its marginals are μ_u, μ_v. Throughout we assume μ and $\mu_u \otimes \mu_v$ are mutually absolutely continuous, and write

$$r(u,v) = \frac{d\mu}{d(\mu_u \otimes \mu_v)}(u,v), \qquad \frac{d\mu_{v|u}(\cdot \mid u)}{d\mu_v}(v) = r(u,v).$$

So $\log r$ is the pointwise mutual information, and $I(u;v) = \mathbb{E}_\mu[\log r] = \mathbb{E}_u\, D_{\mathrm{KL}}(\mu_{v|u} \,\|\, \mu_v)$.

A trained model is a *score* $s : U \times V \to \mathbb{R}$. It defines a joint model and its conditionals (the paper's eqs. (13)–(16)):

$$\nu(du, dv) = \frac{e^{s(u,v)}}{Z}\, \mu_u(du)\mu_v(dv), \qquad \nu_{v|u}(dv \mid u) = \frac{e^{s(u,v)}}{Z(u)}\, \mu_v(dv),$$

$$Z = \iint e^{s}\, d\mu_u\, d\mu_v, \qquad Z(u) = \int e^{s(u,v)}\, \mu_v(dv).$$

Up to constants that do not depend on s, the losses we trained are (the paper's Thm 3.2, Thm 3.8

and Remark 3.12):

$$\begin{aligned} J_{v|u}(s) &= \mathbb{E}_u\, D_{\mathrm{KL}}\big(\mu_{v|u}(\cdot \mid u) \,\|\, \nu_{v|u}(\cdot \mid u)\big) && \text{one-way (forecasting direction),} \\ J_2(s) &= \lambda_u J_{v|u}(s) + \lambda_v J_{u|v}(s), \quad \lambda_u, \lambda_v > 0 && \text{two-way,} \\ J_{\mathrm{joint}}(s) &= D_{\mathrm{KL}}(\mu \,\|\, \nu) && \text{joint.} \end{aligned}$$

The forecast we score is the paper's eq. (31) run on a finite *bank* $v_1, \dots, v_N$ of real futures. This is self-normalised importance sampling with proposal μ_v:

$$\nu^N_{v|u}(\cdot \mid u) = \sum_{j=1}^{N} w_j\, \delta_{v_j}, \qquad w_j = \frac{e^{s(u,v_j)}}{\sum_k e^{s(u,v_k)}}, \qquad \mathrm{ESS} = \Big(\sum_j w_j^2\Big)^{-1}.$$

Score families. With encoders f (past) and g (future) and a scale $\kappa > 0$ (the logit scale), the three families in `test_bed/losses/clip_losses.py` are

$$s_{\mathrm{dot}} = \kappa\, \langle f(u), g(v) \rangle, \qquad s_{\mathrm{ncos}} = \kappa \left\langle \tfrac{f(u)}{|f(u)|}, \tfrac{g(v)}{|g(v)|} \right\rangle, \qquad s_{L^2} = -\tfrac{\kappa}{2}\, |f(u) - g(v)|^2.$$

So $s_{L^2} \le 0$ and $|s_{\mathrm{ncos}}| \le \kappa$, while s_{dot} is unbounded whenever g is.

2 Part I: boundedness decides whether the model exists

Definition 1. A score s is *admissible* for μ (in the forecasting direction) if $Z(u) < \infty$ for μ_u-a.e. u. It is *jointly admissible* if $Z < \infty$.

If s is not admissible, then $\nu_{v|u}(\cdot \mid u)$ does not exist and $J_{v|u}(s) = +\infty$. The probabilistic reading of contrastive learning then has nothing to converge to.

Proposition 2 (Bounded above ⇒ always admissible)**.** *If $s \le c$, then $Z(u) \le e^c$ and $Z \le e^c$ for* every μ. *In particular L^2 ($c = 0$) and normalised cosine ($c = \kappa$) are admissible for every data distribution, with any encoders.*

Proposition 3 (Scores linear in the future need exponential moments)**.** *Let $s(u, v) = \langle a(u), Hv \rangle + b(u)$, which covers the unnormalised dot product with a linear future encoder. Then $Z(u) = e^{b(u)} M_{\mu_v}(H^\top a(u))$, where M_{μ_v} is the moment generating function of μ_v. If μ_v is a multivariate Student-t (any degrees of freedom), then $M_{\mu_v}(w) = \infty$ for every $w \neq 0$. So s is admissible only if $H^\top a(u) = 0$, that is, only if the model ignores the past. Every informative parameter has $J_{v|u} = +\infty$.*

Proposition 4 (ReLU MLPs do not rescue the dot product)**.** *Let g be a ReLU network with biases and let g_∞ be the same network with its biases removed. Then g_∞ is continuous and positively homogeneous, and $\sup_v |g(v) - g_\infty(v)| < \infty$. If $\langle f(u), g_\infty(v_0) \rangle > 0$ for some v_0 and μ_v is an elliptical Student-t, then $Z(u) = \infty$. So the dot product is admissible at u only if $\langle f(u), g_\infty(v) \rangle \le 0$ for all v: the score may not grow in any direction of the future. Nothing in training enforces this. A bounded output layer for g (e.g.* tanh*) does enforce it, by Proposition 2, without normalising anything.*

Admissibility is not just a population-level nicety. It is exactly the condition under which the bank forecast works.

Proposition 5 (Admissible ⇔ the bank forecast is consistent)**.** *Fix u and draw the bank i.i.d. from μ_v. If $Z(u) < \infty$, then $\nu^N_{v|u}(\cdot \mid u) \to \nu_{v|u}(\cdot \mid u)$ weakly, almost surely, and*

$$\frac{\mathrm{ESS}}{N} \to \frac{1}{\mathbb{E}_{\mu_v}\left[(d\nu_{v|u}/d\mu_v)^2\right]} \quad \textit{whenever that second moment is finite.}$$

If $Z(u) = \infty$, the normaliser $\frac{1}{N}\sum_j e^{s(u,v_j)}$ diverges and there is no limit to converge to.

When $Z(u) = \infty$ we can say what happens instead. The forecast collapses onto one future.

Proposition 6 (Collapse under heavy-tailed scores)**.** *Fix u and let $X_j = s(u, v_j)$ be i.i.d. with a regularly varying right tail, $P(X > x) = x^{-\alpha}L(x)$ with $\alpha > 0$ and L slowly varying. Then*

$$\max_j w_j \to 1 \quad \textit{in probability as } N \to \infty.$$

For the linear score of Proposition 3 on Student-t futures with ν degrees of freedom, X is a scaled univariate t, so this holds with $\alpha = \nu$.

So for an unbounded score on heavy tails, *a bigger bank makes the forecast worse.* It supplies more extreme futures to collapse onto. For bounded scores a bigger bank only helps (Proposition 5). Table 1 checks both directions: a linear score with $a = 1$ on unit-variance futures, against an L^2 score on the same Student-t futures. For light tails the limit is explicit: with Gaussian futures and a linear score, $\mathrm{ESS}/N \to e^{-\sigma^2}$ with $\sigma^2 = \mathrm{Var}\, s(u, v)$. Here that is $e^{-1} = 0.368$, as observed.

bank size N	linear score, Gaussian	linear score, Student-t ($\nu = 3$)	L^2 score, Student-t ($\nu = 3$)
10^2	0.081 (0.41)	0.240 (0.30)	0.018 (0.79)
10^3	0.016 (0.38)	0.455 (0.038)	0.002 (0.79)
10^4	0.003 (0.37)	0.760 (0.0007)	$< 10^{-3}$ (0.79)
10^5	0.001 (0.37)	0.902 (10^{-5})	$< 10^{-4}$ (0.79)
10^6	$< 10^{-4}$ (0.37)	0.954 (10^{-6})	$< 10^{-5}$ (0.79)

Table 1: Largest bank weight, mean over 200 banks, with ESS/N in brackets. Only the unbounded score on heavy tails collapses, and it collapses more as the bank grows.

Proposition 7 (Bounded on both sides: guaranteed efficiency, capped sharpness)**.** *If $|s| \le \kappa$ (normalised cosine), then for every u*

$$e^{-2\kappa} \le \frac{d\nu_{v|u}}{d\mu_v} \le e^{2\kappa}, \qquad D_{\mathrm{KL}}(\nu_{v|u} \,\|\, \mu_v) \le 2\kappa, \qquad \frac{\mathrm{ESS}}{N} \to \textit{a limit } \ge e^{-2\kappa}.$$

So normalised cosine has a bank requirement that is uniform in u. The price is that no forecast can be more than $e^{2\kappa}$ times as concentrated as climatology, and none can carry more than 2κ nats of information about the future. The sharpness of the whole model is set by a single number, the logit scale.

What this adds to the write-up's §7. The write-up guessed that $Z(u)$ is a moment generating function and that bounded scores always define a valid model. Propositions 2–3 make that precise. Proposition 4 explains why MLP cosine still failed on heavy tails in goal 2, and Proposition 5 turns validity of the population model into consistency of the forecast we actually score. Proposition 6

	unbounded (dot product)	bounded above (L^2)	bounded both sides (norm. cosine)
model exists	only with light-tailed futures	always	always
bank forecast consistent	only with light-tailed futures	always	always
bank needed	can be infinite (collapse)	finite, $\approx e^{D_{\mathrm{KL}}(\nu_{v\mid u}\|\mu_v)}$ [2], can be large	$\leq$ order $e^{2\kappa}$, uniform in u
sharpness	unlimited	unlimited	capped by κ

Table 2: The boundedness hierarchy.

turns "a few extreme futures take all the weight" into a theorem. One honesty point: real rain is physically bounded, so there $Z(u)$ is finite and Proposition 6 does not apply literally. What applies is its finite-N shadow: $\mathrm{ESS}/N \approx 1/\mathbb{E}[(d\nu/d\mu_v)^2]$ is tiny when the score's distribution over the bank is heavy-tailed. The direct check proposed in the write-up (embedding norm against rain amount, and weight concentration) is still the way to confirm the mechanism on rain.

3 Part II: misspecification decides when the loss variant matters

All three losses aim at the same object, the log density ratio, but they leave different parts of it free.

Proposition 8 (Common target)**.** *For any score s:*

(i) $J_{v|u}(s) = 0 \iff s(u,v) = \log r(u,v) + c(u)$ *for some function c;*

(ii) $J_{u|v}(s) = 0 \iff s(u,v) = \log r(u,v) + c'(v)$ *for some function c';*

(iii) $J_2(s) = 0 \iff J_{\mathrm{joint}}(s) = 0 \iff s = \log r + const.$

Definition 9. A score family $\mathcal{S}$ is *well-specified* for a loss if that loss attains 0 on $\mathcal{S}$. By Proposition 8, it is well-specified for the one-way loss if it contains some $\log r + c(u)$, and for the two-way and joint losses if it contains $\log r + \mathrm{const}$.

Corollary 10 (Agreement under well-specification)**.** *If $\mathcal{S}$ contains $\log r + const$, then the population minimisers of the one-way, the two-way (for any $\lambda_u, \lambda_v > 0$) and the joint loss all give exactly $\nu_{v|u} = \mu_{v|u}$. The choice of loss variant cannot matter at the population level.*

So a difference between loss variants is a symptom of misspecification. The converse is false: P3 (one-way and two-way agree for the dot product) is a case of a misspecified family whose variants agree anyway. The paper's Theorem 5.1 shows that for that family the weighting only rescales the loss.

3.1 The Gaussian case: P1, P3 and P4 as misspecification

Let μ be a centred Gaussian with covariance blocks C_{uu}, C_{uv}, C_{vv}, joint precision P, and $D = \mathrm{blockdiag}(C_{uu}, C_{vv})$. Then

$$\log r(u,v) = -\tfrac{1}{2}u^\top(P_{uu} - C_{uu}^{-1})u \;-\; u^\top P_{uv} v \;-\; \tfrac{1}{2}v^\top(P_{vv} - C_{vv}^{-1})v + \mathrm{const},$$

with $P_{vv} = C_{v|u}^{-1}$ and $P_{uu} = C_{u|v}^{-1}$, so both "self" terms are positive semidefinite. With linear encoders every score is a quadratic,

$$s(u,v) = -\tfrac{1}{2}u^\top R u + u^\top A v - \tfrac{1}{2}v^\top S v,$$

and the families differ only in which (R, A, S) they allow:

family	allowed (R, A, S)	well-specified for
dot product	$R = S = 0$	nothing (if $C_{uv} \neq 0$)
L^2	$\left(\begin{smallmatrix} R & -A \\ -A^\top & S \end{smallmatrix}\right) = [G, -H]^\top [G, -H] \succeq 0$	one-way only
dot + quadratics	R, S symmetric, A free	every loss

- **Dot product (P1).** It has no v-only term, so $\nu_{v|u}$ has covariance C_{vv} whatever A is. The mean can be right, but the spread is stuck at climatology, which is the paper's Corollary 5.2.
- **Dot product, joint (P4).** It is additionally missing the u-only term. And the joint normaliser is finite only if $\|C_{uu}^{1/2} A C_{vv}^{1/2}\| < 1$, which is where the paper's shrinkage function h (Thm 5.6) comes from. Our numerical minimiser reproduces h exactly.
- **L^2, one-way (P2).** It is well-specified: the v-only term $-\frac{1}{2} v^\top S v$ with $S \succeq 0$ can match $P_{vv} - C_{vv}^{-1}$. This is the paper's Theorem 5.3 and Corollary 5.4.

Proposition 11 (L^2 is misspecified for the two-way and joint losses)**.** *Under linear L^2, the model's joint precision is $D^{-1} + Q$ with $Q \succeq 0$, so every model joint covariance satisfies $(D^{-1} + Q)^{-1} \preceq D$. The data covariance satisfies $C \not\preceq D$ whenever $C_{uv} \neq 0$, because $D - C$ has eigenvalues $\pm$ the singular values of C_{uv}. Hence $\log r + \text{const}$ is not in the family, and neither the two-way nor the joint loss can be minimised to zero. In words: an L^2 model can couple u and v only by shrinking both marginals, and real data couple them without shrinking.*

The PSD constraint also ties the two directions together. By a Schur complement, $S \succeq A^\top R^+ A$: a strong past-to-future coupling A forces either a sharp forecast (large S) or a sharp backward conditional (large R). We return to this in §4.

Proposition 12 (Scalar closed form)**.** *For scalar u, v with unit variances and correlation $\rho \in (0, 1)$, the joint-L^2 optimum forecasts*

$$\nu_{v|u}(\cdot \mid u) = \mathcal{N}\Big(\frac{\rho}{2-\rho}\, u,\ \frac{2(1-\rho)}{2-\rho}\Big), \qquad \textit{against the truth} \qquad \mathcal{N}\big(\rho\, u,\ 1-\rho^2\big).$$

The mean is shrunk by a factor $1/(2-\rho)$ and the forecast is over-confident (its variance is too small by a factor $2/(2+\rho-\rho^2)$). The fraction of the available information lost, $J_{v|u}/I(u;v)$, tends to $3/8$ as $\rho \to 0$ and to 0 as $\rho \to 1$.

Table 3 gives the full grid at the population optimum of each loss, as the fraction of $I(u; v)$ that the resulting $v \mid u$ forecast loses. It is computed in closed form from Gaussian KLs, with no training and no sampling.

Three readings of Table 3:

- The dot product and two-way/joint L^2 fail in *opposite regimes*. The dot product's error is in the spread, which is small when dependence is weak and large when it is strong. L^2's two-way/joint error is the coupling-by-shrinking constraint, which bites hardest when dependence is weak. At $\rho = 0.1$, two-way L^2 loses 29% of the information and the dot product loses 0.5%.
- The goal-1 field is in the strong-dependence regime ($I = 1.63$ nats). There, two-way and joint L^2 lose 0.3%. That is why goal 1 measured one-way, two-way and joint L^2 all at +0.0% on the Gaussian field and why the write-up's scorecard gives P3 a tick for L^2. The theory says that tick is regime-dependent.

setting	$I(u;v)$	dot product 1-way	dot product 2-way	dot product joint	L^2 1-way	L^2 2-way	L^2 joint	dot + quad. all three
$\rho = 0.1$	0.005	0.005	0.005	0.005	0	0.294	0.333	0
$\rho = 0.3$	0.047	0.046	0.046	0.051	0	0.219	0.251	0
$\rho = 0.5$	0.144	0.131	0.131	0.157	0	0.148	0.170	0
$\rho = 0.7$	0.337	0.272	0.272	0.323	0	0.081	0.092	0
$\rho = 0.9$	0.830	0.512	0.512	0.571	0	0.021	0.023	0
goal-1 GP, $50 \to 50$	1.627	0.704	0.704	0.746	0	0.003	0.003	0

Table 3: Fraction of the information $I(u;v)$ (nats) lost by the forecast at the population optimum: 0 is perfect, 1 is climatology. The last row uses the exact covariances of the goal-1 Gaussian domain (`test_bed/config.yaml`). Source: `test_bed/results/theory/misspecification.md`.

- Adding free u-only and v-only quadratic terms to the dot product makes every loss exact. This sharpens the write-up's open question 1: a future-only term should fix P1 for the one-way loss, and adding a past-only term as well should make P3 and P4 disappear.

3.2 Nonlinear encoders: well-specified in the bulk, not in the tails

If f and g can output a constant coordinate (biases allow this), then $\langle f(u), g(v)\rangle$ contains terms $c(u) + d(v)$. Sums $\sum_k f_k(u)g_k(v)$ are dense in the continuous functions on compact sets (Stone–Weierstrass). So with enough width, the MLP dot-product family is approximately well-specified *on compacts*. By Corollary 10, the loss variants then converge, and P1 and P4 fade. That is goal 2.

Approximation on compacts says nothing about growth at infinity, which is exactly what Part I is about. The two parts therefore divide the work cleanly:

in the bulk of the data, misspecification decides; in the tails, boundedness decides.

Linear normalised cosine is the remaining misspecified case. It sees the past only through the direction $Au/|Au|$, so it cannot represent a $\log r$ that depends on the level of the past. That explains its +12–23% with linear encoders (write-up, Table 2).

4 Where the theory and the experiments disagree

On weather, the population theory predicts the wrong ranking among the L^2 variants. One-way L^2 is the well-specified one, yet it lost to joint in every case, and to two-way in two of three:

skill (2 seeds)	L^2 1-way	L^2 2-way	L^2 joint
rain, linear (vs climatology)	−0.106	+0.011	+0.012
temperature, MLP (vs persistence)	+0.152	+0.243	+0.280
temperature, linear (vs persistence)	+0.352	+0.344	+0.383

The population argument bounds what joint and two-way can lose: at most about 3/8 (joint) or 1/3 (two-way) of $I(u;v)$ in the scalar Gaussian case. On rain, $I(u;v)$ itself is tiny (the best model beats climatology by 1%), so that bound is negligible in absolute terms. The observed gaps are far larger, so they must be finite-sample effects. A consistent reading:

- One-way L^2 has no check on sharpness. On rain its bank weights collapsed (minimum ESS 1.6 linear, median ESS 3 with MLPs; `summary_B.md`), even though the score is bounded above. A bounded-above score still allows an arbitrarily sharp forecast (Table 2).

- Two-way and joint L^2 do have such a check. Through $S \succeq A^\top R^+ A$, a sharp forecast must come with a sharp backward conditional, which the two-way loss scores. The joint loss also scores the model's marginals, which a collapsed forecast distorts.

This is a bias–variance reading of the write-up's open question 3 ("why is joint consistently best?"). The misspecification is a bias that is small exactly when the signal is weak, and it buys a regulariser against over-sharpening, which is worst exactly then. It predicts that the one-way deficit shrinks as training data grows, and turns into the population ranking in the large-data limit. Goals 1–2, with plentiful synthetic data, are consistent with that: there all L^2 variants tie. This reading is a hypothesis, not a result.

5 Predictions and cheap tests

Each item below is a direct consequence of a proposition above and can be run with the existing test bed.

1. **Bank-size sweep** (Prop. 6). For unnormalised cosine on the df = 3 domain, the largest weight should rise and skill should fall as the bank grows. For L^2 and normalised cosine, the opposite.
2. **Bounded, unnormalised future encoder** (Props. 2, 4). The dot product with a tanh output on g should not collapse on heavy tails. This separates "bounded" from "normalised".
3. **Quadratic-augmented linear dot product** (Table 3, last column). Adding a learned $-\frac{1}{2}v^\top Sv$ should fix P1 one-way. Adding $-\frac{1}{2}u^\top Ru$ as well should make two-way and joint exact, so P4 disappears. This answers open question 1.
4. **Weak-dependence Gaussian** (Prop. 12). On a domain with weak past-future dependence (scalar $\rho \approx 0.3$, or a GP with a gap between past and future), trained on a lot of data, linear two-way/joint L^2 should trail one-way by roughly 20–25% of the information. The energy-score gap will be smaller but should have the same sign.
5. **Data-size sweep on weather** (§4). If the bias–variance reading is right, the gap between one-way and joint L^2 shrinks as training data grows.
6. **Normalised-cosine information cap** (Prop. 7). The learned forecast's $D_{\mathrm{KL}}(\nu_{v|u} \| \mu_v)$, which can be estimated from the bank weights, should never exceed 2κ. Tuning κ after training should control its calibration on its own (open question 2).

6 Scope and caveats

- Everything in Part II is at the population level: the best member of each family under each loss, with infinite data and infinite negatives. Finite-sample statements are limited to Propositions 5 and 6, and to the hypothesis in §4.
- The numbers in Table 3 are information (KL) losses. The experiments report the energy score, which is less sensitive to spread errors. The two rankings should agree in sign, but the sizes are not comparable.
- Table 3 assumes full embedding dimension. The rank-constrained case (the paper's Thms 5.1, 5.3 and 5.6 with $r < \min(n_u, n_v)$) adds a further, separate source of misspecification.

- Proposition 4 covers ReLU-type (piecewise-linear) networks. Other activations need their own growth bound.

A Proofs

Proposition 2. $Z(u) = \int e^s \, d\mu_v \le e^c \mu_v(V) = e^c$, and the same bound holds for Z. For L^2, $-\frac{\kappa}{2}|f-g|^2 \le 0$. For normalised cosine, Cauchy–Schwarz gives $|\langle \bar{f}, \bar{g} \rangle| \le 1$. □

Proposition 3. $Z(u) = e^{b(u)} \mathbb{E} \exp\langle H^\top a(u), v \rangle$ by definition. A multivariate t is elliptical, so for $w \neq 0$ the projection $\langle w, v \rangle$ is a scaled univariate t with ν degrees of freedom. Its density decays like $|x|^{-\nu-1}$, so $\mathbb{E} e^{\langle w, v \rangle} = \infty$. □

Proposition 4. Write $g = W_L \sigma(\dots \sigma(W_1 v + b_1) \dots) + b_L$ with $\sigma = \mathrm{ReLU}$, and g_∞ for the same map with all $b_\ell = 0$. ReLU is 1-Lipschitz and positively homogeneous, so by induction over layers $|g(v) - g_\infty(v)| \le \sum_\ell \|b_\ell\| \prod_{k>\ell} \|W_k\| =: B < \infty$, and $g_\infty(tv) = t g_\infty(v)$ for $t \ge 0$. Let $\delta = \langle f(u), g_\infty(v_0) \rangle > 0$. By continuity there is an open cone $K \ni v_0$ on which $\langle f(u), g_\infty(v/|v|) \rangle \ge \delta/(2|v_0|) =: \delta'$. On K, $s(u,v) \ge \kappa(\delta'|v| - |f(u)|B)$. An elliptical t gives every open cone a tail $\mu_v(K \cap \{|v| > t\}) \asymp t^{-\nu}$, so $\int_K e^{\kappa\delta'|v|} \, d\mu_v = \infty$. □

Proposition 5. For bounded continuous φ, $\frac{1}{N}\sum_j e^{s(u,v_j)} \varphi(v_j) \to \int e^s \varphi \, d\mu_v$ and $\frac{1}{N}\sum_j e^{s(u,v_j)} \to Z(u)$ a.s., by the strong law of large numbers, because both integrands are μ_v-integrable when $Z(u) < \infty$. Taking the ratio gives $\int \varphi \, d\nu^N_{v|u} \to \int \varphi \, d\nu_{v|u}$. Applying the same law to the countable convergence-determining class gives weak convergence a.s. For the ESS, write $\rho = d\nu_{v|u}/d\mu_v = e^s/Z(u)$. Then $N \sum_j w_j^2 = \frac{\frac{1}{N}\sum_j \rho(v_j)^2}{(\frac{1}{N}\sum_j \rho(v_j))^2} \to \mathbb{E}_{\mu_v}[\rho^2]/1$. If $Z(u) = \infty$, the law of large numbers for nonnegative variables gives $\frac{1}{N}\sum_j e^{s(u,v_j)} \to \infty$ a.s. □

Proposition 6. Let $X_{(1)} \ge X_{(2)} \ge \cdots$ be the order statistics and b_N the $1 - 1/N$ quantile of X. Then $b_N = N^{1/\alpha}\ell(N)$ with ℓ slowly varying. By the classical extreme-value limit for regularly varying tails, $(X_{(1)}, X_{(2)})/b_N \Rightarrow (\Gamma_1^{-1/\alpha}, \Gamma_2^{-1/\alpha})$, where $\Gamma_1 < \Gamma_2$ are the first two arrival times of a unit-rate Poisson process. These are a.s. distinct, so for every $\varepsilon > 0$ there is $c > 0$ with $P(X_{(1)} - X_{(2)} \ge c\, b_N) \ge 1 - \varepsilon$ for large N. On that event,

$$1 - \max_j w_j \;\le\; \sum_{j \neq (1)} e^{X_j - X_{(1)}} \;\le\; N\, e^{-c\, b_N} \;\to\; 0,$$

because b_N grows polynomially and so faster than $\log N$. □

Proposition 7. With $|s| \le \kappa$, we have $Z(u) \in [e^{-\kappa}, e^{\kappa}]$, so $d\nu_{v|u}/d\mu_v = e^s/Z(u) \in [e^{-2\kappa}, e^{2\kappa}]$. Then $D_{\mathrm{KL}}(\nu_{v|u} \| \mu_v) = \mathbb{E}_\nu \log(d\nu/d\mu_v) \le 2\kappa$ and $\mathbb{E}_{\mu_v}[(d\nu/d\mu_v)^2] = \mathbb{E}_\nu[d\nu/d\mu_v] \le e^{2\kappa}$, and the ESS limit follows from Proposition 5. □

Proposition 8. (i) $J_{v|u}(s) = 0$ iff $\nu_{v|u}(\cdot \mid u) = \mu_{v|u}(\cdot \mid u)$ for a.e. u. Comparing densities with respect to μ_v, that means $e^{s(u,v)}/Z(u) = r(u,v)$, i.e. $s = \log r + \log Z(u)$. Conversely, any $s = \log r + c(u)$ gives $Z(u) = e^{c(u)} \int r(u,\cdot) \, d\mu_v = e^{c(u)}$, and hence the true conditional. (ii) is symmetric. (iii) With $\lambda_u, \lambda_v > 0$, $J_2 = 0$ iff both (i) and (ii) hold, so $c(u) = c'(v)$ for $\mu_u \otimes \mu_v$-a.e. (u,v). A function of u alone that equals a function of v alone almost everywhere under a product measure is a.e. constant. For the joint loss, $D_{\mathrm{KL}}(\mu \| \nu) = 0$ iff $e^s/Z = r$, i.e. $s = \log r + \log Z$. □

Proposition 11. The joint model has density proportional to $e^s \mathcal{N}(0, D)$. With $s = -\frac{1}{2} z^\top Q z$ and $z = (u, v)$, that is $\mathcal{N}(0, (D^{-1} + Q)^{-1})$. Since $D^{-1} + Q \succeq D^{-1}$, inverting reverses the order and $(D^{-1} + Q)^{-1} \preceq D$. On the data side, $D - C = \begin{pmatrix} 0 & -C_{uv} \\ -C_{uv}^\top & 0 \end{pmatrix}$, whose eigenvalues are $\pm\sigma_i(C_{uv})$. So $C \preceq D$ forces $C_{uv} = 0$. Zero two-way or joint loss requires the model joint to equal μ (Proposition 8(iii)), hence $(D^{-1} + Q)^{-1} = C$, which is impossible when $C_{uv} \neq 0$. The Schur-complement inequality $S \succeq A^\top R^+ A$ is the generalised Schur criterion for $\begin{pmatrix} R & -A \\ -A^\top & S \end{pmatrix} \succeq 0$. □

Proposition 12. With $s = -\frac{1}{2}(gu - hv)^2$ and unit variances, $Q = \begin{pmatrix} g^2 & -gh \\ -gh & h^2 \end{pmatrix}$ and, up to constants, $2J_{\text{joint}} = \operatorname{tr}((I + Q)C) - \log\det(I + Q) = g^2 + h^2 - 2gh\rho - \log(1 + g^2 + h^2)$. For fixed $t = g^2 + h^2$ the cross term is minimised at $g = h$, which leaves $t(1 - \rho) - \log(1 + t)$, minimised at $t = \rho/(1 - \rho)$. The model forecast has precision $1 + h^2 = 1 + t/2 = (2 - \rho)/(2(1 - \rho))$ and mean $gh/(1 + h^2) = \rho/(2 - \rho)$. Expanding $J_{v|u}$ for small ρ gives $J_{v|u} = \frac{3}{16}\rho^2 + O(\rho^3)$, while $I(u; v) = -\frac{1}{2}\log(1 - \rho^2) = \frac{1}{2}\rho^2 + O(\rho^4)$, so the ratio tends to $3/8$. Table 3 confirms the closed form numerically: at $\rho = 0.5$ it gives 0.1699. □

References

[1] R. Baptista, A. Stuart and S. Tran. *A Mathematical Perspective on Contrastive Learning.* arXiv:2505.24134, 2025.

[2] S. Chatterjee and P. Diaconis. The sample size required in importance sampling. *Annals of Applied Probability*, 28(2):1099–1135, 2018.